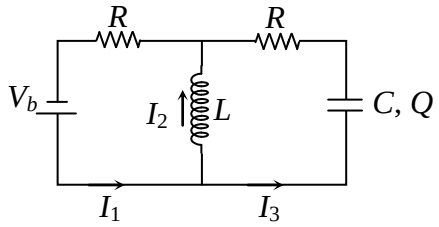
PHYSICS

SECTION – I
Single Correct Choice Type

This section contains 4 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

---

1. In the given the circuit, which of the following choices is a possible system of equations representing the complete Kirchoff's law description of the circuit?



(a) $\begin{cases} V_b = I_1 R \\ Q = CV_b \\ V_b = -L\dfrac{dI_2}{dt} \end{cases}$

(b) $\begin{cases} V_b - L\dfrac{dI_2}{dt} - RI_1 = 0 \\ -\dfrac{Q}{C} - R(I_1 - I_2) + L\dfrac{dI_2}{dt} = 0 \\ I_3 = I_1 - I_2 \end{cases}$

(c) $\begin{cases} V_b - L\dfrac{dI_2}{dt} - RI_1 = 0 \\ -QC - R(I_1 + I_2) + L\dfrac{dI_2}{dt} = 0 \end{cases}$

(d) $\begin{cases} V_b - L\dfrac{dI_2}{dt} + RI_1 = 0 \\ -\dfrac{Q}{C} - RI_3 - L\dfrac{dI_2}{dt} = 0 \\ I_1 = I_2 + I_3 \end{cases}$

2. A steady non-turbulent stream of water comes out of a tap and falls vertically downward. As it does so, the diameter of the stream appears to get smaller. The primary reason will be

(a) The water's surface tension constricts the stream

(b) Air pressure, which decreases with altitude, squeezes the stream

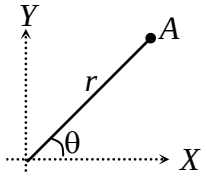
(c) The water is accelerating under gravity and so the stream must get thinner as the flow rate (velocity multiplied by cross-sectional area) must be constant.

(d) The flow does not constrict; it is an optical illusion.

3. Relative density of a material is found by ratio of weight in air and decrease in weight in water. If weight of a body in air and water are measured as $w_a = (10.0 \pm 0.1)\,\text{g}$ and $w_w = (5.0 \pm 0.1)\,\text{g}$, then percentage error in relative density will be

(a) 5 (b) 10 (c) 2.5 (d) 7.5

4. A short dipole is oriented along x-axis such that direction of dipole moment is along positive x-axis. It is observed that the magnitude of electric field and electric potential are equal at a point $\sqrt{5}\,\text{m}$ from the centre of the dipole as shown. The value of θ is



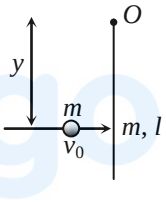
(a) 30^0 (b) 37^0 (c) 53^0 (d) 45^0

SECTION – II
Multiple Correct Choice Type

This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

---

5. A uniform rod of mass m and length l is free to rotate about a fixed horizontal axis O. A small ball of mass m moving horizontally with velocity v_0 strikes the rod at depth y from O as shown in figure. If angular velocity of rod and velocity of ball just after the collision, be ω and v respectively, then the wrong statements will be



(a) If $y = l/2$, then in case of elastic collision, $v = \frac{v_0}{2}$ and $\omega = \frac{v_0}{l}$

(b) If $y = \frac{l}{2}$, then in case of elastic collision, momentum of ball-rod system remains conserved.

(c) If momentum of ball-rod system remains conserved, then $y > \frac{l}{2}$.

(d) Ball cannot be rebounded back whatever is the value of y.

6. In an African wildlife reserve a jeep is traveling at a constant speed of 105 km/hr. Alongside a cheetah is running at constant speed. The animal, frightened by the Jeep's noise, keeps a constant distance of 60m from the vehicle. Not having paid much attention to the surrounding landscape, the driver suddenly realize that the jeep is passing the very same spot where it was 60 seconds before. Then

(a) speed of cheetah is 105.4 km/hr

(b) speed of cheetah is 82.4 km/hr

(c) distance travelled by cheetah is 1373.2 m

(d) distance travelled by cheetah is 1750m

7. An ideal inductor of inductance L = 10H carrying current 5A is connected to a resistor of resistance 20Ω and mass 2 kg and specific heat 0.5 J $kg^{-1}K^{-1}$. Assume that all energy given to resistor is used to increase its temperature. Then

(a) The total change in temperature of resistor will be 125°C.

(b) The total change in temperature of resistor will be 250°C.

(c) The total charge passed through the resistor is 2.5Columb.

(d) The total charge passed through the resistor is infinite.

8. An object is taken from a point P to another point Q in a gravitational field. Then

(a) assuming the earth to be spherical, if both P and Q lie on earth's surface the work done is zero

(b) if P is on earth's surface and Q above it, the work done is minimum when it is taken along the straight line PQ

(c) the work done depends only on the position of P and Q and is independent of the path along which the particle is taken

(d) there is no net work done if the object is taken from P to Q and then brought back to P, along any path

SECTION – III
Reasoning Type

This section contains 4 reasoning type questions. Each question contains Statement-1 and Statement-2. Each question has 4 choices (a), (b), (c) and (d), out of which **ONLY ONE** is correct.

Directions: Read the following questions and choose

(A) **Statement-1** is True, **Statement-2** is True; **Statement-2** is a correct explanation for **Statement-1**.

(B) **Statement-1** is True, **Statement-2** is True; **Statement-2** is not a correct explanation for **Statement-1**.

(C) **Statement-1** is True, **Statement-2** is False.

(D) **Statement-1** is False, **Statement-2** is True.

9. **Statement-1:** In a conductor of non-uniform cross-section, the current is same, though the drift velocity is different.

Statement-2: In series current is same and for same current drift velocity is inversely proportional to cross-sectional area $\left(v_d \propto \frac{1}{A}\right)$.

(a) A (b) B (c) C (d) D

10. **Statement-1:** Two conductors are identical in shape and size but are of different material. If both conductors are placed in an identical electric field, charges induced on them may be different.

Statement-2: Induced charge $q' = -q\left(1 - \frac{1}{K}\right)$; more the value of K, more will be the induced charge.

(a) A (b) B (c) C (d) D

11. **Statement-1:** If a particle moves in xy plane it has only z-component of angular momentum.

Statement-2: Angular momentum $\vec{L} = \vec{r} \times \vec{p}$ is always perpendicular to position vector $\vec{r} = (\hat{i}x + \hat{j}y)$.

(a) A (b) B (c) C (d) D

mathongo

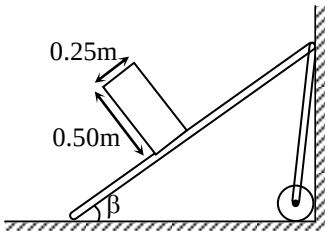
12. **Statement-1:** A race car traveling around circular track have a non zero impulse.
Statement-2: The impulse is zero only when there is no net change in momentum.
(a) (A) (b) (B) (c) (C) (d) (D)

SECTION–IV
Comprehension Type

This section contains 2 groups of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

Passage-I

An engineer is designing a conveyor system for loading lay bales into a wagon. Each bale is 0.25m wide, 0.50 m high, and 0.80 m long (the dimension perpendicular to the plane of the figure), with mass 30.0 kg. The centre of gravity of each bale is at its geometrical centre. The coefficient of static friction between a bale and the conveyor belt is 0.60, and the belt moves with constant speed. The angle β of the conveyor is slowly increased. At some critical angle (β_1) a bale will tip (if it doesn't slip first), and at some different critical angle (β_2) it will slip (if it doesn't tip first)



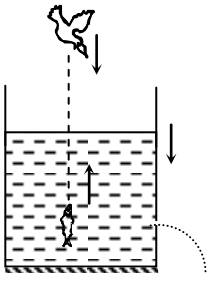
13. The first critical angle, β_1 (in the same conditions) at which it tips without slipping is
(a) $\tan^{-1}0.50$ (b) $\tan^{-1}0.60$
(c) $\tan^{-1}0.40$ (d) $\tan^{-1}0.20$

14. The second critical angle, β_2 (in the same conditions) at which it slips without tipping is
(a) $\tan^{-1}0.50$ (b) $\tan^{-1}0.60$
(c) $\tan^{-1}0.12$ (d) $\tan^{-1}0.70$

15. The first critical angle, β_1 at which it tips if the coefficient of friction were 0.40 is
(a) $\tan^{-1}0.50$ (b) $\tan^{-1}0.60$
(c) $\tan^{-1}0.70$ (d) $\tan^{-1}0.20$

Passage-II

A bird in air is diving vertically over a tank with speed 5 cm/s. Base of tank is silvered. A fish in the tank is rising upward along the same line with speed 2 cm/s. Water level is falling at rate of 2 cm/s. [Take $\mu_{water} = 4/3$]



16. Speed of the image of fish as seen by the bird directly in cm/s is
(a) 8 (b) 6 (c) 3 (d) 4

17. Speed of the image of fish formed after reflection in the mirror as seen by the bird in cm/s is
(a) 8 (b) 6 (c) 3 (d) 4

18. Speed of image of bird relative to the fish looking upwards in cm/s is
(a) 8 (b) 6 (c) 3 (d) 4

SECTION–V
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

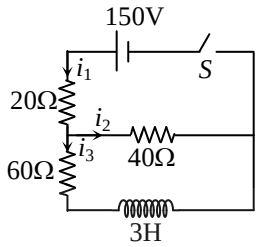
If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.

	1	2	3	4	5
A	●	○	○	●	●
B	○	●	●	○	○
C	●	●	○	○	○
D	○	○	○	●	●

1. The position (x) of a particle varies with time as $x = t^3 - 3t^2$, where t is time in second. Match column I with column II.

Column I	Column II
(A) Position of particle is zero at	**1.** $t = 0$ s
(B) Velocity of particle is zero at	**2.** $t = 1$s
(C) Magnitude of acceleration of particle is zero at	**3.** $t = 2$ s
(D) Magnitude of displacement is equal to distance traveled by the particle at	**4.** $t = 3$ s **5.** t = 1.5 s

2. Match the values of i_1, i_2, i_3 for the circuit shown under conditions of column I with column II.



Column -I	Column -II
(A) Soon after switch is closed	**1.** i_1 = 0 A
(B) Long after switch is closed	**2.** i_2 = 1.37 A
(C) Long after switch is reopened	**3.** i_3 = 1.37 A
(D) Soon after switch is reopened	**4.** $i_1 = i_2$ **5.** $i_2 = i_3$

CHEMISTRY

SECTION – I

Single Correct Choice Type

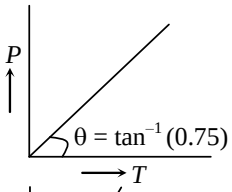
This section contains 8 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

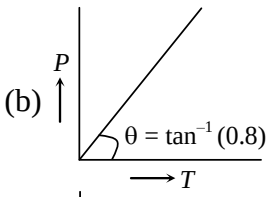
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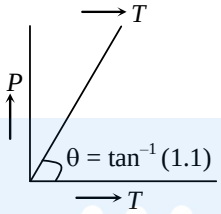
1. 1 mol of gas A dissociates in a closed rigid container of volume 0.16 lit as per the following reaction:

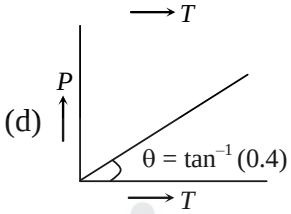
$$2A(g) \longrightarrow 3B(g) + 2C(g)$$

If α for A is 0.4 irrespective of temperature, then the correct P(atm) vs T(K) graph is: [Use R = 0.08 lit at m/mol k)

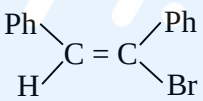
(a) 

(b) 

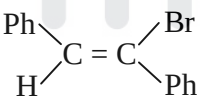
(c) 

(d) 

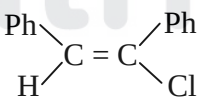
2. Arrange the following compounds according to decreasing rate of elimination with NaOEt/EtOH.



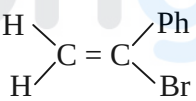
(A)



(B)



(C)



(D)

(a) (A) > (B) > (C) > (D) (b) (B) > (A) > (C) > (D)
(c) (D) > (A) > (C) > (B) (d) (B) > (D) > (A) > (C)

3. 4 g of an unknown monoprotic weak acid is dissolved in water. This solution required 50 mL of 0.4 M NaOH for complete neutralization. When the same solution of monoprotic acid is titrated with 25 mL of given NaOH solution. The pH of resulting solution is found to be 4. The pK_a of the weak acid is

(a) 2.5 (b) 3.0 (c) 3.5 (d) 4.0

4. $(X) + C + Cl_2 \xrightarrow{1000\,K} (Y) + CO$

$(Y) + 2H_2O \rightarrow Be(OH)_2 + 2HCl$

Compound (Y) exist in polymeric chain structure and it is an electron deficient molecule. The compound (Y) is

(a) BeO (b) $BeO.Be(OH)_2$

(c) $Be(OH)_2$ (d) $BeCl_2$

5. The conductivity of a saturated solution of CaF_2 at 18°C was found to be 4.2×10^{-5} S cm^{-1} and the conductivity of water used for making the solution was 2×10^{-6} S cm^{-1}. The molar ionic conductivities at infinite dilution of Ca^{2+} and F^- are 104 and 48 S cm^2 mol^{-1}, respectively. The solubility product of CaF_2 will be

(a) 9.84×10^{-12} M^3 (b) 3.2×10^{-11} M^3

(c) 3.63×10^{-11} M^3 (d) 3.7×10^{-11} M^3

6. Identify the correct sequence of increasing number of π-bonds in the structures of the following molecules:

(I) $H_2S_2O_6$ (II) H_2SO_3 (III) $H_2S_2O_5$

(a) (I), (II) and (III) (b) (II), (III), (I)

(c) (II), (I) and (III) (d) (I), (III) and (II)

7. The Henry's law constant for the solubility of N_2 gas in water at 298 K is 1×10^{-5} atm^{-1}. The percentage of N_2 in air is 80%. The number of moles of N_2 from air that dissolved in 10 moles of water at 298 K and 5 atm pressure is

(a) 4×10^{-5} (b) 4×10^{-4}

(c) 5×10^{-4} (d) 5×10^{-5}

8. A sample containing H_3PO_4 and P_4O_{10} is labelled in the analogous manner as oleum. The mass of P_4O_{10} in a sample labelled as 127%, would be (atomic masses: O = 16, P = 31)

(a) 71 g (b) 27 g

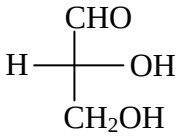
(c) 83 g (d) 35 g

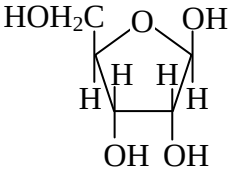
SECTION – II
Multiple Correct Choice Type

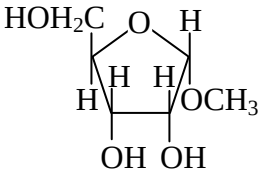
This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

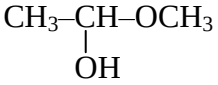
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9. Which of the following undergo aqueous Br_2 oxidation?

(a) 

(b) 

(c) 

(d) 

10. Identify the incorrect statement(s)

(a) Solvay process cannot be used to prepare K_2CO_3 due to precipitation of $KHCO_3$

(b) The gas evolved during commercial preparation of NaOH using Castner-Kellner cell is oxygen and H_2

(c) All alkaline earth metal halides are ionic in nature

(d) The hemihydrate of calcium sulphate is known as gypsum

11. The equilibrium between three gaseous isomers A, B and C can be represented as:

$$A(g) \rightleftharpoons B(g) \quad K_1 = ?$$
$$B(g) \rightleftharpoons C(g) \quad K_2 = 0.4$$
$$C(g) \rightleftharpoons A(g) \quad K_3 = 0.6$$

If initially one mole of A is taken in a closed vessel of volume 1 litre, then which of the following is/are correct statements?

(a) [A] + [B] + [C] = 1 M at any time during the reaction.

(b) Concentration of [C] is $\frac{1}{2.1}$ M at equilibrium.

(c) $K_1 = \frac{1}{0.24}$

(d) Isomer (A) is least stable thermodynamically.

---

12. Which of the following is/are correct about $C_3H_4Cl_2$ (A) ?

(a) (A) has total number of isomers as 11.

(b) All isomers of (A) are reducible and give addition products.

(c) All isomers of (A) can undergo free radical substitution reactions.

(d) All isomers of (A) cannot give electrophilic substitution reactions.

SECTION–III
Comprehension Type

This section contains 2 groups of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

Passage-I

Metallic gold is frequently found in aluminosilicate rocks and it is finely dispersed among other minerals. It may be extracted by treating the crushed rock with aerated sodium cyanide solution. During this process, metallic gold is slowly converted to $[Au(CN)_2]^{2-}$, which is water soluble, the metallic gold is recovered from it by reacting the gold complex with zinc, which is converted to $[Zn(CN)_4]^{2-}$.

Gold in nature is frequently alloyed with silver, which is also oxidised by aerated sodium cyanide solution, similar to gold.

500 L of a solution of 0.01 M $[Au(CN)_2]^-$ and 0.003 M $[Ag(CN)_2]^-$ was evaporated to one third of the original volume and was treated with 40 g zinc. Assuming that deviation from standard conditions is unimportant in this case and that all these redox reactions go essentially to completion. Given:

$$[Zn(CN)_4]^{2-} + 2e^- \longrightarrow Zn + 4CN^- \quad ; \quad E° = -1.26\ V$$

$$[Au(CN)_2]^- + e^- \longrightarrow Au + 2CN^- \quad ; \quad E° = -0.60\ V$$

$$[Ag(CN)_2]^- + e^- \longrightarrow Ag + 2CN^- \quad ; \quad E° = -0.31\ V$$

13. The concentrations of $[Ag(CN)_2]^-$ after the addition of Zn in it, would be

(a) 0.03 M (b) 0.003 M

(c) 0.0018 M (d) 0.018 M

14. What concentration of cyanide ion is required to keep 99 mol % of the gold in solution in the form of the cyanide complex? Given: K_f of $[Au(CN)_2]^-$ is 4×10^{28}.

(a) 2×10^{14} M (b) 5×10^{-14} M

(c) 2×10^{-14} M (d) 5×10^{14} M

15. There have been several efforts to develop alternative gold extraction processes, which would replace the present one. The reason for this is

(a) NaCN solution corrodes mining machinery.

(b) gold obtained by this process is impure.

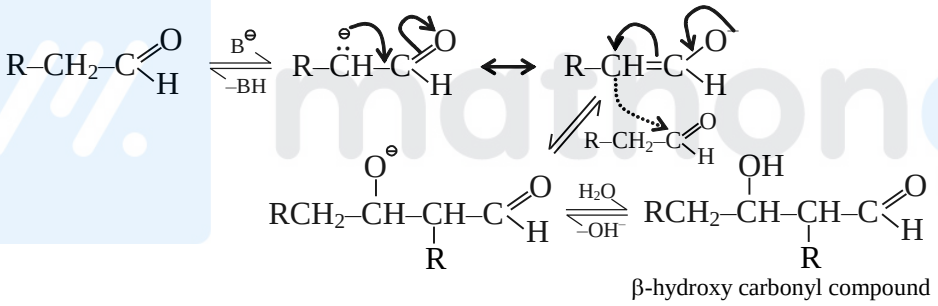
(c) silver is precipitated faster than gold.

(d) NaCN escapes into ground water and releases HCN.

Passage-II

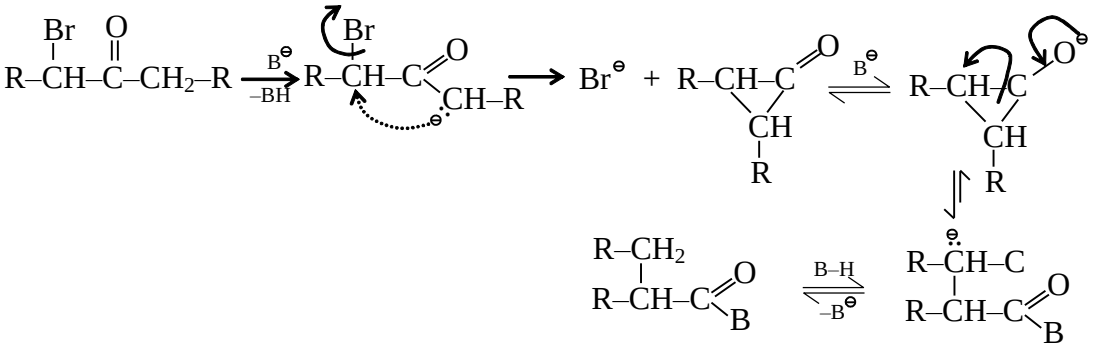
OH^-/OR^- can act as base as well as nucleophile in its reactions. In its reactions with carbonyl compounds (aldehydes, ketones, esters) it act as base and give various condensation and rearrangement reactions.

It undergoes aldol type reaction (OH^- act as base in such reactions) by the following route.

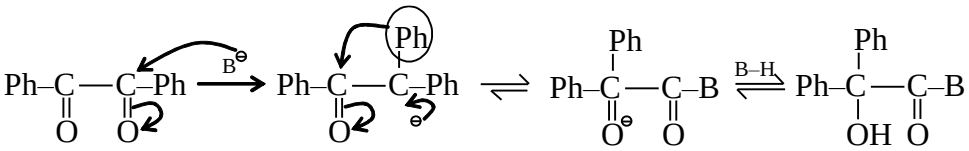


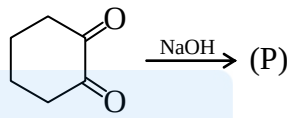
β-hydroxy carbonyl compound

It also undergoes favorski type reactions (OH^-/OR^- act as base in such reaction) by the following pathway.

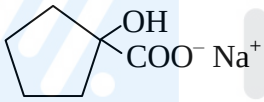


It can also undergo Benzil-Benzilic acid rearrangement (OH^- act as a nucleophilie) as per the given mechanism.

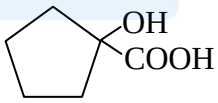


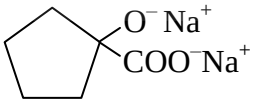
16.  (P)

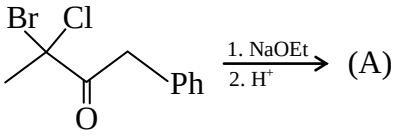
The product (P) is

(a) 

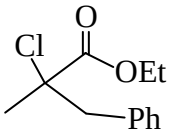
(b) 

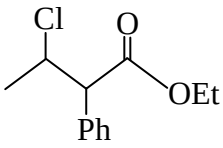
(c) 

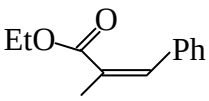
(d) 

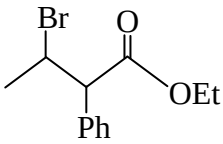
17. 

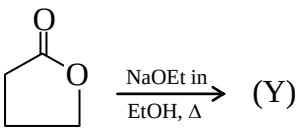
The product (A) in the given reaction would be

(a) 

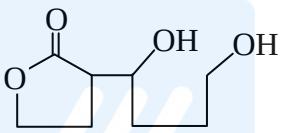
(b) 

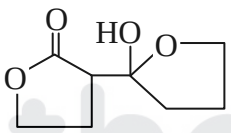
(c) 

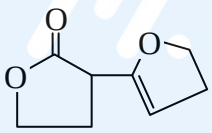
(d) 

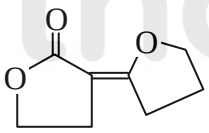
18. 

The product (Y) of the given reaction is

(a) 

(b) 

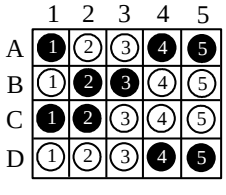
(c) 

(d) 

SECTION–IV
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.

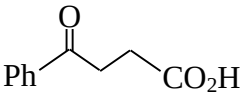
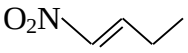
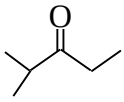


1.

Column I	Column II
(A) $Sb^{3+} \rightarrow (SbO)^{+}$	**1.** Acidification
(B) $CrO_4^{2-} \rightarrow Cr_2O_7^{2-}$	**2.** Addition of base
(C) $MnO_4^{-} \rightarrow MnO_4^{2-}$	**3.** Hydrolysis
(D) $B_4O_7^{2-} \rightarrow B(OH)_3$	**4.** Colour change
	5. Reduction

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2.

Column I (Compounds)	Column II (Oxidising/reducing agents)
(A) R–CH=N–R	**1.** SeO_2
(B) 	**2.** Zn–Hg/HCl
(C) 	**3.** $NaBH_4$
(D) 	**4.** H_2–$[RhCl(Ph_3P)_3]$
	5. CF_3CO_3H

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MATHEMATICS

SECTION – I
Single Correct Choice Type

This section contains 9 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

---

1. Let $f(x)=\int_0^x \sin\left(t^2-t+x\right)dt$, then value of $f''\left(-\frac{1}{2}\right)+f\left(-\frac{1}{2}\right)$ is

(a) 0 (b) π (c) $\frac{\pi}{2}$ (d) $\frac{3\pi}{2}$

2. Lines $(1+\lambda)x+(4-\lambda)y+(2+\lambda)=0$ and $(4-\lambda)x-(1+\lambda)y+(6-3\lambda)=0$ are concurrent at points *A* and *B* respectively and intersect at *C*, then locus of centroid of ΔABC is

(a) $\left(x+\frac{3}{2}\right)^2+\left(y+\frac{7}{10}\right)^2=\frac{34}{225}$ (b) $\left(x-\frac{3}{2}\right)^2+\left(y-\frac{7}{10}\right)^2=\frac{34}{225}$

(c) $\left(x-\frac{3}{2}\right)^2+\left(y+\frac{7}{10}\right)^2=\frac{34}{225}$ (d) none of these

3. $f:[-1,1]\to R$ such that $f(0)=f(1)$ and $\forall\, a,b\in[-1,1]\ :\ |f(b)-f(a)|<|b-a|$, then which of the following is always true?

(a) $\exists c\in(-1,1): f'(c)>1$ (b) $|f(b)-f(a)|<1/2$

(c) $f(x)\le x$ (d) $f(b)\ne f(a)\ \forall a\ne b$.

4. If 84^n divides $(496)!$, then the maximum value of *n* can be

(a) 160 (b) 40 (c) 80 (d) none of these

5. The fundamental period of function $f(x)=\cos^7 x+\cos^7\left(x+\frac{2\pi}{3}\right)+\cos^7\left(x+\frac{4\pi}{3}\right)$ is

(a) 2π (b) $\frac{2\pi}{3}$ (c) $\frac{4\pi}{3}$ (d) $\frac{\pi}{3}$

6. One side of square *ABCD* is on the line $y = 2x - 17$ and other two vertices are on parabola $y = x^2$, then the minimum area of square is

(a) 80 (b) $4\sqrt{5}$ (c) 1280 (d) $16\sqrt{5}$

7. The length of the perpendicular from the origin to the plane passing through three non-collinear points $\vec{a}, \vec{b}, \vec{c}$ is

(a) $\dfrac{[\vec{a}\,\vec{b}\,\vec{c}]}{|\vec{a}\times\vec{b}+\vec{c}\times\vec{a}+\vec{b}\times\vec{c}|}$ (b) $\dfrac{2[\vec{a}\,\vec{b}\,\vec{c}]}{|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}|}$

(c) $[\vec{a}\,\vec{b}\,\vec{c}]$ (d) none of these

8. Let O be an interior point of ΔABC such that $\overrightarrow{OA}+2\overrightarrow{OB}+3\overrightarrow{OC}=\vec{0}$, then the ratio of area of ΔABC to area of ΔAOC is

(a) 2 (b) $\dfrac{3}{2}$ (c) 3 (d) $\dfrac{5}{3}$

9. The number of 5 digit multiples of 3 that end with digit 6 is

(a) 30000 (b) 6000 (c) 2999 (d) 3000

SECTION – II
Multiple Correct Choice Type

This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

---

10. A drawer contains red and black balls. When two balls are drawn at random, the probability that they both are red is $\frac{1}{2}$. The number of balls in the drawer can be

(a) 21 (b) 11 (c) 4 (d) 3

11. If a function satisfies $(x-y)f(x+y)-(x+y)f(x-y)=2(x^2y-y^3)\ \forall\ x, y \in R$ and $f(1)=2$, then

(a) $f(x)$ must be polynomial function (b) $f(3)=12$

(c) $f(0)=0$ (d) $f(x)$ may not be differentiable

12. $f:[1,4]\to[7,14]$ is a surjective, twice differentiable function such that $f''(x)>0$, $\forall x\in(1,\ 4)$, then the equation $(f'(x))^2=\frac{49}{9}$ has

(a) at least one root in (1, 4) (b) exactly one root in (1, 4)
(c) at least two roots in (1, 4) (d) at most two roots in (1, 4)

13. In a ΔABC medians AD and BE are perpendicular to each other, then

(a) $\frac{1}{2}<\frac{a}{b}<2$ (b) $\frac{1}{2}<\frac{c}{a}<2$

(c) $\frac{1}{2}<\frac{a}{c}<2$ (d) $\frac{1}{2}<\frac{b}{a}<2$

SECTION – III
Reasoning Type

This section contains 3 reasoning type questions numbered 14 to 16. Each question contains Statement-1 and Statement-2. Each question has 4 choices (a), (b), (c) and (d), out of which **ONLY ONE** is correct.

Directions: Read the following questions and choose

(A) **Statement-1** is True, **Statement-2** is True; **Statement-2** is a correct explanation for **Statement-1**.
(B) **Statement-1** is True, **Statement-2** is True; **Statement-2** is not a correct explanation for **Statement-1**.
(C) **Statement-1** is True, **Statement-2** is False.
(D) **Statement-1** is False, **Statement-2** is True.

14. **Statement-1:** If two points lie on opposite sides of all lines forming a triangle then one of these points has to lie inside the triangle.
Statement-2: If two points lie on opposite side of two intersecting lines then they are in vertically opposite regions.

(a) A (b) B (c) C (d) D

15. **Statement-1 :** If $A=[a_{ij}]_{n\times n}$ and $a_{ij}=i^2-j^2$ then A is necessarily a skew symmetric matrix $n>1$.

Statement-2 : A skew-symmetric matrix of odd order is always invertible.

(a) A (b) B (c) C (d) D

16. **Statement 1 :** If $\sin^{-1}\left(1-y^2\right)+\sin^{-1}\left(1-x\right)=\cos^{-1}\left(1+x^2+y^2+2xy^2\right)$, then locus of $(x,\ y)$ is parabola.

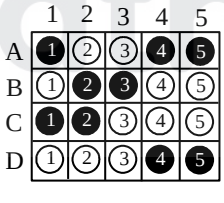
Statement 2 : $\sqrt{1-x-2y}+\sqrt{1-x+2y}=2$ represents a parabola.

(a) A (b) B (c) C (d) D

SECTION–IV
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.



1. The tangent to $y^2 = 4ax$ at $P(at^2, 2at)$ intersects axis at T and the normal at P intersects at N, S is the focus and M is foot of perpendicular from P on directrix.

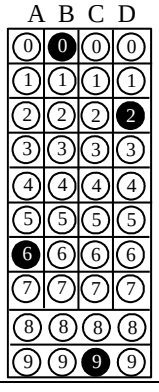
Column -I	Column -II
(A) $a(1+t^2)$	**1.** intercept on normal made by circle with PS as diameter.
(B) $a\sqrt{1+t^2}$	**2.** distance of M from tangent at P.
(C) $\frac{1}{t}$	**3.** distance PN.
(D) $\frac{a}{2}\left(1+t^2\right)$	**4.** $\tan(\angle MPT)$
	5. slope of tangent at P.

2. Let (x, y) be such that $\sin^{-1}(ax)+\cos^{-1}(y)+\cos^{-1}(bxy)=\frac{\pi}{2}$.

Column I	Column II
(A) If $a = 1$ and $b = 0$, then (x, y)	**1.** lies on the circle $x^2+y^2=1$
(B) If $a = 1$ and $b = 1$, then (x, y)	**2.** lies on $\left(x^2-1\right)\left(y^2-1\right)=0$
(C) If $a = 1$ and $b = 2$, then (x, y)	**3.** lies on $4x^2+y^2=1$
(D) If $a = 2$ and $b = 0$, then (x, y)	**4.** lies on the curve having bounds area $\frac{\pi}{2}$
	5. lies on the curve having bounds area π

SECTION–V
Integer Answer Type

This section contains 4 questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers A, B, C and D (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the following:



---

1. Let $\begin{vmatrix} 1+x & x & x^2 \\ x & 1+x & x^2 \\ x^2 & x & 1+x \end{vmatrix} = \frac{1}{6}(x-\alpha_1)(x-\alpha_2)(x-\alpha_3)(x-\alpha_4)$ be an identity in x, where $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ are independent of x. Then find the value of $\alpha_1\alpha_2\alpha_3\alpha_4$.

Space for work

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2. $f:R\to R$ be a twice differentiable function satisfying $f''(x)-5f'(x)+6f(x)\geq 0\ \forall\ x\geq 0$ if $f(0)=1\quad f'(0)=0$. If $f(x)$ satisfies $f(x)\geq ah(bx)-bh(ax),\ \forall x\geq 0$, then find $(a+b)h(0)$.

Space for work

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3. A sequence is obtained by deleting all perfect squares from set of natural numbers. Find the remainder when the 2003rd term of new sequence is divided by 2048.

Space for work

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4. Find the least integral value of a such that $\sqrt{9-a^2+2ax-x^2} > \sqrt{16-x^2}$ for at least one positive x.

Space for work

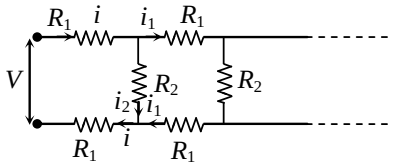
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PAPER-I
PHYSICS
SECTION – I
Single Correct Choice Type

1. $i = i_1 + i_2$ …(1)

$\frac{V}{n} = 2V_{R_1} + \frac{V}{n^2}$ …(2)

$V = 2V'_{R_1} + \frac{V}{n}$ …(3)



$V_{R_1} = \frac{V}{2n^2}(n-1)$

$V'_{R_1} = \frac{V(n-1)}{2n}$

Using equation (1) $\frac{V}{2nR_1}(n-1) = \frac{V(n-1)}{2n^2R_1} + \frac{V}{nR_2}$

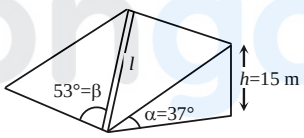
On solving $\frac{R_1}{R_2} = \frac{(n-1)^2}{2n}$

∴ **(b)**

2. $l = \frac{25}{12}h$ …(1)

$a = (g\sin 37^\circ \cos 37^\circ) - (\mu g \cos 37^\circ + \mu g \sin^2 37^\circ)$

$a = \frac{62}{25}$ m/s^2 …(2)



Use $l = \frac{1}{2}at^2$

$t = 5$ sec.

∴ **(a)**

3. For a body to float average density of body should be less than or equal to density of liquid.

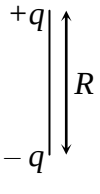
∴ **(d)**

4. $\vec{\tau} = I\vec{\alpha}$

$qRE - fR = \frac{2}{3}mR^2\alpha$ …(1)

$f = ma$ …(2)

$a = R\alpha$ …(3)



On solving above equations,

$f = \frac{3}{5}qE$

∴ **(b)**

5. After a long time the velocity of both the conductors become same.

∴ **(b)**

6. Using conservation of flux

$$L_1\frac{2E}{R} = L_1 i + L_2 i$$

$$i = \frac{L_1\dfrac{2E}{R}}{L_1 + L_2}$$

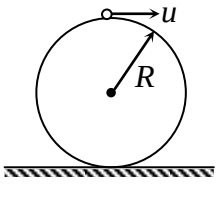
$\therefore$ **(a)**

7. Volume of cube = $(8)^3 = 512\ \text{cm}^3$

As the result should have one significant digit so it should be $500\ \text{cm}^3$.

$\therefore$ **(b)**

8. $mg - N = \dfrac{mu^2}{R}$



If $N \to 0 \Rightarrow u \to \sqrt{gR}$

So range is $0 \le u < \sqrt{gR}$

$\therefore$ **(d)**

SECTION – II
Multiple Correct Choice Type

9. For isochoric process $\dfrac{P}{T}$ = Constant

For adiabatic process $P^{1-\gamma}T^{\gamma}$ = Constant

$$\frac{dP}{dT} = \frac{\gamma}{(\gamma-1)}\left(\frac{P}{T}\right)$$

$\therefore$ **(b, c)**

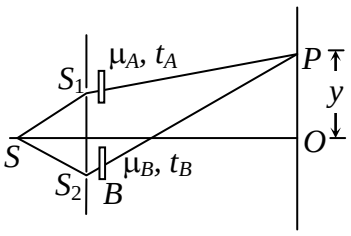
10. Path difference = 0 for central maxima

$$\{SS_2 + (S_2P - t_B) + \mu_B t_B\} - \{SS_1 + (S_1P - t_A) + \mu_A t_A\} = 0$$

$$(S_2P - S_1P) = -\mu_B t_B + t_B - t_A + \mu_A t_A$$

$$y\frac{d}{D} = (\mu_A t_A - \mu_B t_B) + (t_B - t_A)$$

Shift, $y = \dfrac{D}{d}\left[(\mu_A t_A - \mu_B t_B) + (t_B - t_A)\right]$



(i) if $t_B = t_A \Rightarrow y = 0$

(ii) if $t_B > t_A \Rightarrow y > 0$

(iii) if $t_B < t_A \Rightarrow y < 0$

$\therefore$ **(a, b, c)**

11. $E_{K\alpha} = \frac{3}{4}RhcZ^2 = \frac{3}{4}x$, where $x = RhcZ^2$

$E_{L\beta} = \frac{3}{16}x$

$E_{K\beta} = \frac{8}{9}x$

$E_{m\alpha} = \frac{7}{144}x$

$E_{m\gamma} = \frac{15}{16}x$

Now, $E_{k\alpha} + E_{L\beta} = \frac{15}{16}x$

$E_{k\beta} + E_{m\alpha} = \frac{15}{16}x$

$\therefore$ **(a, c, d)**

12. $\frac{1}{y} + \frac{1}{x} = \frac{1}{f}$

$-\frac{1}{y^2}V_i - \frac{1}{x^2}V_0 = 0$

x-direction $V_i = -\left(\frac{y}{x}\right)^2 V_0 = -\left(\frac{-60}{-60}\right)^2(-2\hat{i}) = 2\hat{i}$ cm/s

$m = -\frac{(-60)}{(-60)} = \frac{y'}{x'} \Rightarrow \frac{dy'}{dt} = \hat{j}$ cm/s

$\therefore$ **(a, d)**

SECTION–III
Comprehension Type
Passage – 1

13. $2mV_0 = 3mV \Rightarrow V = \frac{2V_0}{3}$

Impulse = $2m\left(\frac{2V_0}{3}\right) - 2mV_0 = -\frac{2}{3}mV_0$

$\therefore$ **(b)**

14. $KE_{\min} = \frac{1}{2} \times 2m\left(\frac{2}{3}V_0\right)^2 + \frac{1}{2}m\left(\frac{2}{3}V_0\right)^2 = \frac{2}{3}mV_0^2$

$\therefore$ **(a)**

15. $-1=\frac{V_1-V_2}{V_0-0}$ …(1)

$2mV_0=2mV_1+mV_2$ …(2)

$V_1=\frac{V_0}{3};\ V_2=\frac{4}{3}V_0$

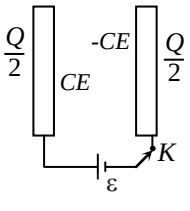
$$\frac{K_1}{K_2}=\frac{\frac{1}{2}2m\left(\frac{V_0}{3}\right)^2}{\frac{1}{2}m\left(\frac{4}{3}V_0\right)^2}=\frac{1}{8}$$

$\therefore$ **(d)**

Passage – 2

16. Charge flown $=\left(CE+\frac{Q}{2}\right)-Q$

$=\left(CE-\frac{Q}{2}\right)$



$\therefore$ **(d)**

17. Charge on the inner surface of left plate should be CE

$\therefore$ **(d)**

18. Work done = $\Delta qE=\left(\frac{3Q}{2}\right)\left(\frac{2Q}{C}\right)=\frac{3Q^2}{C}$

$\therefore$ **(b)**

SECTION–IV
Matrix–Match Type

1. **(A) – (1), (2), (4); (B) – (1), (3), (5); (C) – (2), (4); (D) – (5).**

2. **(A) – (2), (5); (B) – (1), (5); (C) – (3), (4); (D) – (3), (4).**

CHEMISTRY

SECTION – I

Single Correct Choice Type

1. **(b)** $2A(g) \rightarrow 3B(g) + 2C(g)$

$1-\alpha \qquad \frac{3\alpha}{2} \qquad \alpha$

$$\text{Slope} = \frac{n_T R}{V} = \frac{\left(1-\alpha+\frac{3\alpha}{2}+\alpha\right)\times 0.08}{0.16} = \frac{\left(1+\frac{3}{2}\times 0.4\right)}{2} = 0.8$$

2. **(b)** This is because H and Br are trans in position in compound (B). Among (A) and (C), (A) is a vinyl bromide while (C) is a vinyl chloride, so bromide undergoes reaction faster than chloride. (D) reacts slowly because the H is least acidic in it.

3. **(d)** $\frac{W_{acid}}{M_{acid}} = 50\times 0.4\times 10^{-3}$

$$\therefore \quad M_{acid} = \frac{W_{acid}}{50\times 0.4\times 10^{-3}} = \frac{4\times 1000}{50\times 0.4} = 200 \text{ g mol}^{-1}$$

$$\therefore \text{ Mol of acid} = \frac{4}{200} = \frac{1}{50} = 2\times 10^{-2}$$

Mole of base used in titration = $0.40 \times 25 \times 10^{-3} = 10^{-2}$

$\therefore$ Mole of salt formed = 1×10^{-2}

Mole of acid left = 1×10^{-2}

$$\therefore \quad pH = pK_a + \log\frac{S}{A} = pK_a = 4 \quad \left(pH = pK_a + \log\frac{S}{A}\right)$$

4. **(d)** $\underset{(X)}{BeO} + C + Cl_2 \xrightarrow{1000\ K} \underset{(Y)}{BeCl_2} + CO$

$\underset{(Y)}{BeCl_2} + 2H_2O \rightarrow Be(OH)_2 + 2HCl$

5. **(b)** $\kappa_{soln} = \kappa_{CaF_2} + \kappa_{water}$

$\kappa_{CaF_2} = \kappa_{soln} - \kappa_{water} = 4.2 \times 10^{-5} - 0.2 \times 10^{-5} = 4 \times 10^{-5}$ S cm^{-1}

$\underset{a-s}{CaF_2(s)} \rightleftharpoons \underset{s}{Ca^{2+}(aq)} + \underset{2s}{2F^-(aq)}$

$K_{sp} = 4s^3$, $[F^-] = 2[Ca^{2+}]$

$\kappa_{CaF_2} = \kappa_{Ca^{2+}} + \kappa_{F^-}$

$$\kappa_{CaF_2} = \frac{\Lambda_{m,Ca^{2+}}[Ca^{2+}]}{1000} + \frac{\Lambda_{m,F^-}[F^-]}{1000}$$

$$\kappa_{CaF_2} = \frac{\Lambda_{m,Ca^{2+}}[Ca^{2+}]}{1000} + \frac{\Lambda_{m,F^-}2[Ca^{2+}]}{1000} = \frac{[Ca^{2+}]}{1000}\left(\Lambda_{m,Ca^{2+}} + 2\Lambda_{m,F^-}\right)$$

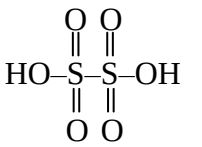
$4 \times 10^{-5} \times 1000 = [Ca^{2+}]\ \ (\Lambda^{\infty}_{m,\,Ca^{2+}} + 2\Lambda^{\infty}_{m,\,F^-}) = [Ca^{2+}]\,(104 + 2 \times 48)$

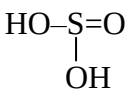
$4 \times 10^{-2} = [Ca^{2+}]\,(200)$

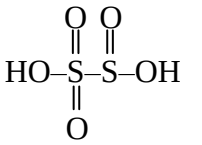
$[Ca^{2+}] = s = \dfrac{4 \times 10^{-2}}{200} = 2 \times 10^{-4}\ M$

$\therefore\ K_{sp} = 4s^3 = 4(2 \times 10^{-4})^3 = 32 \times 10^{-12} = \mathbf{3.2 \times 10^{-11}\ M^3}$.

6. **(b**

$H_2S_2O_6$  4π bonds

H_2SO_3  1π bond

$H_2S_2O_5$  3π bond

7. **(b)** From the unit of Henry's law constant for N_2 gas, it implies that

$$X_{N_2} = k_{N_2} p_{N_2}$$

$$X_{N_2} = 1 \times 10^{-5} \times 5 \times \frac{80}{100} = 4 \times 10^{-5}$$

$$\therefore\ \frac{n_{N_2}}{n_{N_2} + n_{H_2O}} = 4 \times 10^{-5}$$

Since the mole fraction of dissolved N_2 is very small, so $n_{N_2} + n_{H_2O} \simeq n_{H_2O}$

$$\therefore\ \frac{n_{N_2}}{n_{H_2O}} = 4 \times 10^{-5}$$

$n_{N_2} = 4 \times 10^{-5} \times 10 = \mathbf{4 \times 10^{-4}\ moles}$.

8. **(a)** $P_4O_{10} + 6\ H_2O \rightarrow 4H_3PO_4$

Mass of water added is 27 g.

Moles of H_2O added = $\dfrac{27}{18} = \dfrac{3}{2}$

Moles of P_4O_{10} that react with $\dfrac{3}{2}$ moles of H_2O = $\dfrac{3}{2} \times \dfrac{1}{6} = \dfrac{1}{4}$

$\therefore$ Mass of P_4O_{10} in the sample = $\dfrac{1}{4} \times 284 = \mathbf{71\ g}$.

SECTION – II
Multiple Correct Choice Type

9. (a, b, d)

Aldehydic linkage is oxidized by Br_2 water. Hemiacetal dissociate in aqueous solution to give free –CHO group.

10. (a, b, c, d)

11. (a, c, d) $[A] + [B] + [C] = 1$ M

$$\therefore \quad K_3[C] + \frac{[C]}{K_2} + [C] = 1$$

$$\text{or,} \quad [C]\left\{0.6 + \frac{1}{0.4} + 1\right\} = 1$$

$$\text{or,} \quad [C] = \frac{1}{4.1}\text{M}$$

$$\text{Besides, } [C] = K_2[B] = \frac{[A]}{K_3}$$

$$\therefore \quad K_2 \times K_3 = \frac{[A]}{[B]}$$

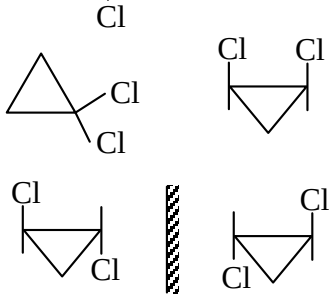
$$\text{or} \quad 0.4 \times 0.6 = \frac{1}{K_1} \quad \Rightarrow \quad K_1 = \frac{1}{0.24}$$

12. (a, b, c) A has 11 isomers including stereoisomers.

$CH_3–CH=CCl_2$ $ClCH_2–CH=CHCl$ (geometrical isomerism)

$Cl_2CH–CH=CH_2$ $ClCH_2–\underset{\displaystyle Cl}{\underset{|}{CH}}=CH_2$

$CH_3–\underset{\displaystyle Cl}{\underset{|}{C}}=CHCl$ (geometrical isomerism)



All isomers are reducible, give addition products and can undergo free radical substitution reaction.

SECTION–III
Comprehension Type

Passage–I

$$2[Ag(CN)_2]^- + Zn \longrightarrow 2Ag + [Zn(CN)_4]^{2-} \; ; \; E_1^o = -0.31 - (-1.26) = 0.95\text{V}$$

$$2[Au(CN)_2]^- + Zn \longrightarrow 2Au + [Zn(CN)_4]^{2-} \; ; \; E_2^o = -0.60 - (-1.26) = 0.66\text{V}$$

$E_1^o > E_2^o$, thus Zn reduces $[Ag(CN)_2]^-$ in preference to $[Au(CN)_2]^-$.

13. (c) Moles of $[Ag(CN)_2]^-$ in 500 L solution = 500 × 0.003 = 1.5

Moles of $[Au(CN)_2]^-$ in 500 L solution = 500 × 0.01 = 5

Moles of Zn added = $\frac{40}{65.4} = 0.61$

According to reaction, 0.61 mole of Zn will consume 1.2 mole of $[Ag(CN)_2]^-$, thus moles of $[Ag(CN)_2]^-$ left = 1.5 – 1.2 = 0.3.

Conc. of $[Ag(CN)_2]^-$ after the reaction = $\frac{0.3}{500/3} = 0.0018$ M.

Conc. of $[Au(CN)_2]^-$ after the reaction = 0.01 × 3 = 0.03 M.

14. (b) $Au^+(aq) + 2CN^-(aq) \rightleftharpoons [Au(CN)_2]^-(aq)$

$$K_f = \frac{[Au(CN)_2]^-}{[Au^+][CN^-]^2} = 4 \times 10^{28}$$

$$\frac{[Au(CN)_2]^-}{[Au^+]+[Au(CN)_2]^-} \times 100 = 99,\ [Au(CN)_2]^- = 99[Au^+]$$

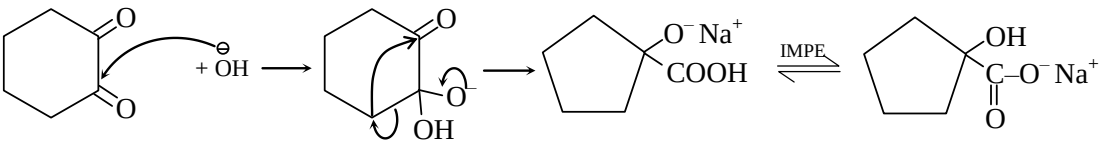
$$\therefore\ K_f = \frac{99[Au^+]}{[Au^+][CN^-]^2}\ ;\ \frac{99}{[CN^-]^2} = 4 \times 10^{28}$$

$$[CN^-] = \sqrt{\frac{99}{4\times10^{28}}} = 4.97 \times 10^{-14} \simeq \mathbf{5 \times 10^{-14}\ M}.$$

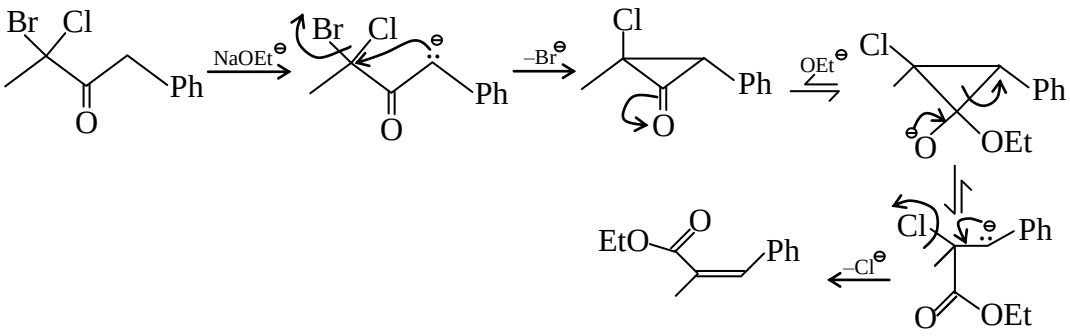
15. (d) This process of gold extraction is not suitable as CN^- passes into the ground water and releases toxic HCN in it, making the human life at high level of risk.

Passage–II

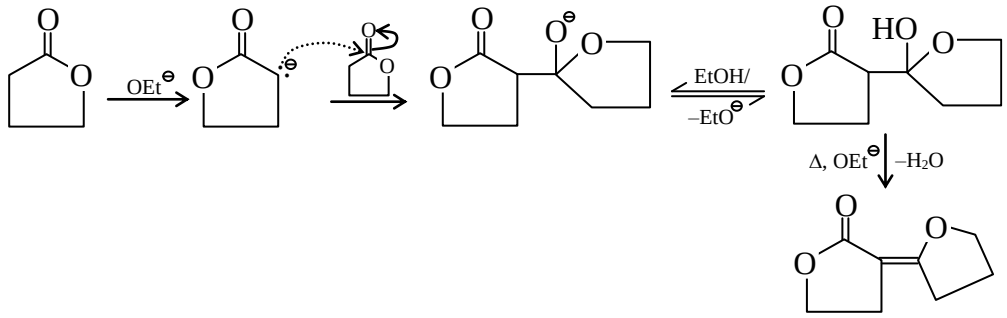
16. (a)



17. (c)



18. (d)



SECTION–IV
Matrix–Match Type

1. **(A) – 3, 4** ; $SbCl_3 + H_2O \longrightarrow \underset{\text{orange ppt.}}{SbO^+Cl^-\downarrow} + 2HCl$
(colourless)

The reaction involves hydrolysis and colour change.

(B) – 1, 4 ; $2H^+ + 2CrO_4^{2-} \rightleftharpoons Cr_2O_7^{2-} + H_2O$
(yellow) (orange)

The equilibrium can be shifted in forward direction by decrease of pH (i.e. on acidification).

(C) – 2, 4, 5 ; $2KMnO_4 + 2KOH \longrightarrow 2K_2MnO_4 + H_2O + O$
(purple) (green)

The reaction occurs by addition of a base, it involves reduction of Mn from +7 to +6 oxidation state and colour changes from purple to green.

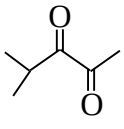
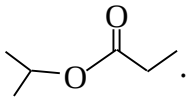
(D) – 1, 3 ; $Na_2B_4O_7 + 2HCl + 5H_2O \longrightarrow 2NaCl + 4H_3BO_3$

This reaction is occuring in presence of a mineral acid and it is a hydrolysis reaction.

2. **(A) – 3** ; Imines can be reduced by $NaBH_4$ readily.

(B) – 1, 2, 3, 5 ; SeO_2 will oxidise it into 1,2-diketone, Zn-Hg/HCl will selectively reduce keto functionality to CH_2, $NaBH_4$ will selectively reduce $\rangle C{=}O$ to $\rangle CH{-}OH$ grouping. CF_3CO_3H will oxidise it via Baeyer-Villiger oxidation.

(C) – 1, 4 ; SeO_2 oxidises alkene at allylic position and introduce OH on allylic carbon. Hydrogenation by Wilkinson's catalyst can reduce double or triple bonds selectively without affecting $-NO_2$, $\rangle C{=}O$, $-CO_2H$ groups etc.

(D) – 1, 2, 3, 5 ; SeO_2 oxidises it to  . Clemenson and $NaBH_4$ reduction for this compound is also feasible. CF_3CO_3H will cause Baeyer-Villiger oxidation to give  .

MATHEMATICS

SECTION – I

Single Correct Choice Type

1. **(a)** $f(x)=\cos x\int_0^x \sin(t^2-t)dt+\sin x\int_0^x \cos(t^2-t)dt$

$f''(x)=-\sin x\sin(x^2-x)+(2x-1)\cos x\cos(x^2-x)$

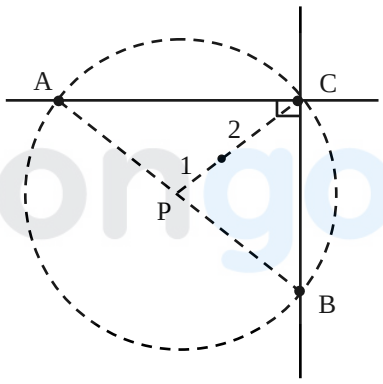
$-\sin x\sin(x^2-x)-\cos x\int_0^x \sin(t^2-t)\,dt+\cos x\cos(x^2-x)$

$+(1-2x)\sin x\sin(x^2-x)-\sin x\int_0^x \sin(t^2-t)dt+\cos x\sin(x^2-x),$

Then $f(x)+f''(x)=(2x+1)\cos x^2$

$f\left(-\frac{1}{2}\right)+f''\left(-\frac{1}{2}\right)=0$

2. **(a)** As lines are perpendicular to each other 'C' moves on a circle with AB as diameter.
Now P, mid-point of AB (which is fixed) when joined with C is median.
$\Rightarrow$ Centroid is moving at a constant distance $\frac{1}{3}(PA)$ from P.
$\Rightarrow$ Locus is a circle



A is point of intersection of $x+4y+2=0$ and $x-y+1=0$ i.e., $\left(-\frac{6}{5},-\frac{1}{5}\right)$

B is point of intersection of $4x-y+6=0$ and $x+y+3=0$ i.e., $\left(\frac{-9}{5},-\frac{6}{5}\right)$

$\Rightarrow P=\left(\frac{-3}{2},\frac{-7}{10}\right)$ therefore locus is $\left(x+\frac{3}{2}\right)^2+\left(y+\frac{7}{10}\right)^2=\frac{34}{225}$

3. **(b)** Consider $a<b$, then $|f(b)-f(a)|<b-a$ …(i)

Also $f(0)=f(1)$, therefore

$|f(b)-f(a)|\le|f(b)-f(0)|+|f(1)-f(a)|$ …(ii)

$|f(b)-f(0)|+|f(1)-f(a)|<1-a+b$ ….(iii)

Adding (i), (ii) and (iii), we get

$|f(b)-f(a)|\le\frac{1}{2}$

4. **(c)** $84=2^2\times3\times7$

$496! = 2^{\lambda}\, 3^{\beta}\, 5^{\gamma}\, 7^{\delta} \ldots\ldots$

Here $\beta > \delta$ and $\lambda > \delta$ as the frequency of occur of 2 is many a times more than that of 3 and 7.

Therefore divisibility will be decided by δ and $\delta = \left[\frac{496}{7}\right] + \left[\frac{496}{49}\right] + \ldots.$ (where [.] greatest integer function)

$= 70 + 10 = 80$

5. **(b)** By inspection $\frac{2\pi}{3}$ is the fundamental period.

Alternately:

If $z = \cos x + i \sin x$; $\frac{1}{z} = \cos x - i \sin x$, we get $128\cos^7 x = \left(z + \frac{1}{z}\right)^7$

$\Rightarrow \quad 64\cos^7 = \cos 7x + 7\cos 5x + 21\cos 3x + 35\cos x$

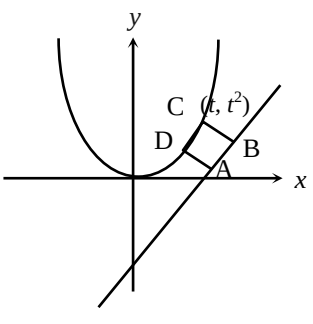
Adding after putting x as $\left(x + \frac{2\pi}{3}\right)$ and $\left(x + \frac{2\pi}{3}\right)$

$64\left(\cos^7 x + \cos^7\left(x + \frac{2\pi}{3}\right) + \cos^7\left(x + \frac{4\pi}{3}\right)\right)$, we get $63\cos 3x$

$\cos^7 x + \cos^7\left(x + \frac{2\pi}{3}\right) + \cos^7\left(x + \frac{4\pi}{3}\right) = \frac{63}{64}\cos(3x)$

Hence period is $\frac{2\pi}{3}$

6. **(a)** $BC = \frac{\left|2t - t^2 - 17\right|}{\sqrt{5}} = a$



$\Rightarrow \quad t^2 - 2t + 17 - \sqrt{5}a = 0 \quad \ldots(i)$

If $C(t_1, t_1^2)$, $D(t_2, t_2^2)$, then

t_1, t_2 are roots of equation (i) and $CD = a$

$\Rightarrow \quad t_1 + t_2 = 2, \quad t_1 t_2 = 17 - \sqrt{5}a$

$\Rightarrow \quad a^2 = (t_1 - t_2)^2 + (t_1^2 - t_2^2)^2$

$a^2 = 4 - 4(17 - \sqrt{5}a) + 4(4 - 4(17 - \sqrt{5}a))$

$a^2 = 20 - 20(17) + 20\sqrt{5}a$

$\Rightarrow \quad a^2 - 20\sqrt{5}a + 320 = 0 \quad \Rightarrow \quad a = 16\sqrt{5}$ or $a = 4\sqrt{5}$

Minimum area is when $a = 4\sqrt{5} \quad \Rightarrow \quad a^2 = 80$

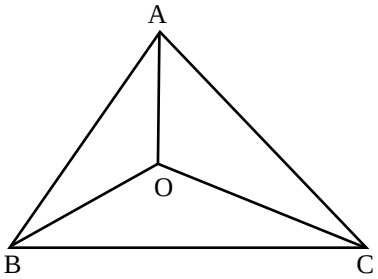
7. **(a)** The vector equation of the plane passing through points $\vec{a}, \vec{b}, \vec{c}$ is

$r.\left(\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\right)=[\vec{a}\ \vec{b}\ \vec{c}]$

Therefore, the length of the perpendicular from the origin to this plane is given by

$$\frac{[\vec{a}\ \vec{b}\ \vec{c}]}{|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}|}.$$

8. **(c)**



$$\frac{\text{area of }(\Delta ABC)}{\text{area of }(\Delta OAC)}$$

$$=\frac{\frac{1}{2}\left|\overrightarrow{OA}\times\overrightarrow{OB}\right|+\left|\overrightarrow{OB}\times\overrightarrow{OC}\right|+\left|\overrightarrow{OC}\times\overrightarrow{OA}\right|}{\frac{1}{2}\left|\overrightarrow{OA}\times\overrightarrow{OC}\right|}$$

$$=\frac{\left|\overrightarrow{OA}\times\overrightarrow{OB}\right|+\left|\overrightarrow{OB}\times\overrightarrow{OC}\right|}{\left|\overrightarrow{OA}\times\overrightarrow{OC}\right|}+1$$

Given $\overrightarrow{OA}+2\overrightarrow{OB}+3\overrightarrow{OC}=\vec{0}$

$2\overrightarrow{OB}\times\overrightarrow{OA}+3\overrightarrow{OC}\times\overrightarrow{OA}=\vec{0}$

$$\Rightarrow \frac{\left|\overrightarrow{OB}\times\overrightarrow{OA}\right|}{3}=\frac{\left|\overrightarrow{OC}\times\overrightarrow{OA}\right|}{2}=\lambda \qquad \ldots(i)$$

$\overrightarrow{OB}\times\overrightarrow{OA}+3\overrightarrow{OB}\times\overrightarrow{OC}=\vec{0}$

$$\Rightarrow \frac{\left|\overrightarrow{OB}\times\overrightarrow{OA}\right|}{3}=\frac{\left|\overrightarrow{OB}\times\overrightarrow{OC}\right|}{1}=\lambda \qquad \ldots(ii)$$

$\Rightarrow$ Ratio of area of Δs i.e., $\frac{\Delta ABC}{\Delta AOC}=\frac{3\lambda+\lambda}{2\lambda}+1=3$

9. **(d)** Total five digit numbers are 90000.

Every third term is divisible by 3 therefore numbers divisible by 3 are 30000.

The multiples of 3 end periodically with digits 3, 6, 9, 2, 5, 8, 1, 4, 7 and 0.

Hence numbers that end with digit 6 are 3000.

Alternately:

Let *abcd* 6 be the five digit numbers.

Therefore the four digit numbers '*abcd*' should be divisible by 3 i.e., $9000\div 3$ i.e, 3000.

SECTION – II
Multiple Correct Choice Type

10. (a), (c)

Let the drawer contains p balls of which 'm' are red.

Probability of drawing two red balls at random is $\frac{{}^{m}C_2}{{}^{p}C_2}=\frac{1}{2}$

$\Rightarrow \quad 2m(m-1)=p(p-1)$

$\Rightarrow \quad 2m^2-2m-p^2+p=0$

$\Rightarrow \quad m=\frac{2\pm\sqrt{4-8(p-p^2)}}{4}=\frac{1\pm\sqrt{1-2p+2p^2}}{2}$

$\Rightarrow \quad 1-2p+2p^2$ should be an odd perfect square.

i.e., $p=21,\ 4$ but $p\neq 3$

when 3 balls out of 4 are red.

15 balls out of 21 are red.

11. (a), (b), (c)

$(x-y)f(x+y)-(x+y)f(x-y)=2y(x-y(x+y))$

Let $x-y=u;\ x+y=v$

$uf(v)-vf(u)=2uv(v-u)$

$\frac{f(v)}{v}-\frac{f(u)}{u}=v-u$

$\left(\frac{f(v)}{v}-v\right)=\left(\frac{f(u)}{u}-u\right)$=constant

Let $\frac{f(x)}{x}-x=\lambda$

$\Rightarrow \quad f(x)=(\lambda x+x^2)$

$f(1)=2$

$\lambda+1=2 \quad \Rightarrow \quad \lambda=1$

$f(x)=x^2+x$

12. (a), (d)

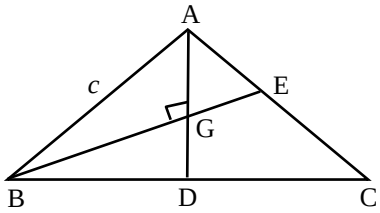
Since $f''(x)>0$ graph of $f(x)$ is concave upward.

Now $f'(x)=\pm\frac{7}{3}$ having a solution would mean line having slope $\pm\frac{7}{3}$ touching the curve.

The graph will strictly lie in the rectangle formed by $x=1,\ x=4,\ y=7,\ y=14$.

If f is monotonic the line parallel to either of diagonal of rectangle will touch the graph exactly once and if f is non-monotonic, then lines parallel to diagonals will touch the graph exactly once.

13. (a), (d)



$AG = \frac{1}{3}\sqrt{2b^2 + 2c^2 - a^2}$ and

$BG = \frac{1}{3}\sqrt{2a^2 + 2c^2 - b^2}$

$AG^2 + BG^2 = c^2$

$\Rightarrow \quad (2b^2 + 2c^2 - a^2) + (2a^2 + 2c^2 - b^2) = 9c^2$

$\Rightarrow \quad a^2 + b^2 = 5c^2 \quad$ ...(i)

and for ΔABC exist

$a - b < c$ or $b - a < c$

$\Rightarrow (a-b)^2 < c^2 \Rightarrow a^2 + b^2 - 2ab < \frac{a^2 + b^2}{5} \Rightarrow 2a^2 + 2b^2 - 5ab < 0$

$\Rightarrow 2\left(\frac{a}{b}\right)^2 - 5\left(\frac{a}{b}\right) + 2 < 0 \Rightarrow \frac{a}{b} = \frac{5 \pm \sqrt{25-16}}{4} = \frac{5 \pm 3}{4} = 2, \frac{1}{2}$

$\Rightarrow \frac{1}{2} < \frac{a}{b} < 2$

Similarly, $\frac{1}{2} < \frac{b}{a} < 2$

SECTION – III
Reasoning Type

14. (d)

15. (c) $a_{ij} = i^2 - j^2 \Rightarrow a_{ji} = j^2 - i^2 = -a_{ij}$

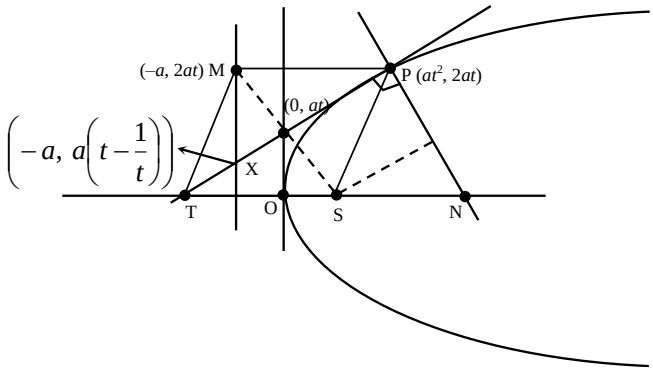
Thus 'A' is necessarily a skew symmetric matrix.

Determinant of skew-symmetric matrix of odd order is always zero.

16. (d)

SECTION–IV
Matrix–Match Type

1. A-1, 3; B-2; C-5; D-4



2. From given equation $\cos^{-1}(y)+\cos^{-1}(bxy)=\frac{\pi}{2}-\sin^{-1}(ax)$

Consider $\cos^{-1} y=\alpha, \cos^{-1}(bxy)=\beta, \cos^{-1}(ax)=\gamma$

$\Rightarrow \alpha+\beta=\gamma \Rightarrow \beta=\gamma-\alpha$ also $\cos\alpha = y, \cos\beta = bxy, \cos\gamma = ax$

$\Rightarrow \cos(\gamma-\alpha)=bxy$

$\Rightarrow \cos\alpha\cos\gamma+\sin\gamma\sin\alpha=bxy \Rightarrow (axy-bxy)^2=\sin^2\gamma\sin^2\alpha=(1-\cos^2\gamma)(1-\cos^2\alpha)$

$\Rightarrow (a-b)^2x^2y^2=(1-a^2x^2)(1-y^2)$

(A)- 1, 5 If $a = 1, b = 0$, then $x^2+y^2=1$

(B)- 2 If $a = 1, b = 1$, then $(x^2-1)(y^2-1)=0$

(C)- 1, 5 If $a = 1, b = 2$, then $x^2+y^2=1$

(D)- 3, 4 If $a = 2, b = 0$, then $4x^2+y^2=1$

SECTION–V
Integer Answer Type

1. Since it is an identity the value of L.H.S. and R.H.S. are equal for all values of x

Put $x=0 \Rightarrow \alpha_1\alpha_2\alpha_3\alpha_4=6$

Ans. 6

2. Given inequality can be written as:

$f''(x)-2f'(x)\geq 3(f'(x)-2f(x))$

Let $f'(x)-2f(x)=g(x)$

$\Rightarrow g'(x)-3g(x)\geq 0$ Multiply e^{-3x}

$\Rightarrow \left(g(x)e^{-3x}\right)'\geq 0 \Rightarrow g(x)e^{-3x}$ is non-decreasing.

Now $g(0)=f'(0)-2f(0)=-2$

$g(x)e^{-3x}\geq -2, \forall x\geq 0$

$f'(x)-2f(x)\geq -2e^{3x}, \forall x\geq 0$ Multiply e^{-2x}

$\Rightarrow \left(f(x)e^{-2x}\right)'\geq -2e^{x}, \forall x\geq 0$

$\Rightarrow \left(f(x)e^{-2x}+2e^{x}\right)'\geq 0$

$\Rightarrow f(x)e^{-2x}+2e^{x}\geq 3$

$\Rightarrow f(x)\geq 3e^{2x}-2e^{3x}, \forall x\geq 0$

Comparing $ah(bx)-bh(ax)$ with $3e^{2x}-2e^{3x}$ we get $h(x)=e^x, a=3, b=2$

$\Rightarrow (a+b)h(0)=5$

Ans. 5

3. Since $[\sqrt{2046}=[\sqrt{2047}]=[\sqrt{2048}]=[\sqrt{2049}=45$

$\therefore$ 2003rd term is $2003+45=2048$

Hence remainder is 0

Ans. 0

4. $y=\sqrt{9-a^2+2ax-x^2}$ $\Rightarrow$

$(x-a)^2+y^2=9$

For given inequality to hold for positive x.

$a-3<4$

$\Rightarrow$ $a<7$ $\Rightarrow$ $a=6$

Ans. 6

